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Faculty of Prince Al-Hussein Bin Abdallah II For Information Technology
Information Technology Department

AI-Powered Healthy Indoor Environment

A project submitted
in partial fulfillment of the requirements for the
B.Sc. Degree in Data science and Artificial Intelligence

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2025/2026

CERTIFICATE

It is hereby certified that the project titled

< ***AI-Powered Healthy Indoor Environment***>

, submitted by undersigned, in partial fulfillment of the award of the degree of "Bachelor in Data science and Artificial Intelligence" embodies original work done by them under my supervision.

All the analysis, design and system development have been accomplished by the undersigned. Moreover, this project has not been submitted to any other college or university.

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ABSTRACT

Indoor air quality (IAQ) has a direct impact on human health, particularly for individuals suffering from respiratory conditions such as asthma or sensitivity to enclosed spaces. Poor indoor air conditions caused by high carbon dioxide levels, inadequate ventilation, and unfavorable thermal conditions can lead to discomfort and serious health risks.

This project proposes an intelligent system that predicts indoor air quality using machine learning techniques based on environmental and contextual features, including carbon dioxide concentration, temperature, relative humidity, occupancy level, and window status. The IAQ index is used as a reference standard to label air quality levels into predefined categories. Several machine learning and deep learning models are trained to learn the relationship between these factors and the resulting air quality condition.

Based on the predicted air quality level, the system provides real-time, user-centered recommendations to improve indoor conditions, such as increasing ventilation or reducing occupancy. The proposed approach demonstrates how artificial intelligence can be effectively applied to enhance indoor environmental quality and support health-aware decision-making in enclosed spaces.

ACKNOWLEDGEMENTS

We would like to express our deepest gratitude to our project supervisor, Dr. Nadera Aljawabreh, for her invaluable guidance, continuous support, and constructive feedback throughout the duration of this project.

We also extend our sincere thanks to our instructors and the Department of Information Technology (IT) for providing us with the resources, knowledge, and opportunities necessary to carry out this research.

To our classmates and colleagues, we thank you for the insightful discussions, collaborative spirit, and mutual support that made this journey both enriching and enjoyable.

A special note of appreciation goes to our friends for their constant moral support and motivation. Your encouragement has been a driving force throughout this project.

Lastly, we are profoundly grateful to our families for their unwavering support, patience, and understanding. Their belief in our abilities and continuous encouragement have been the foundation of our academic journey.

Thank you to everyone who contributed to making this project a reality.

TABLE OF CONTENTS

ABBREVIATIONS

AI – Artificial Intelligence

ML – Machine Learning

DL – Deep Learning

IAQ – Indoor Air Quality

AQI – Air Quality Index

CO_2 – Carbon Dioxide

CO – Carbon Monoxide

RH – Relative Humidity

VOC – Volatile Organic Compounds

PM – Particulate Matter

WHO – World Health Organization

ASHRAE – American Society of Heating, Refrigerating and Air-Conditioning Engineers

IT – Information Technology

CNN – Convolutional Neural Network

LSTM – Long Short-Term Memory

KNN – K-Nearest Neighbors

SVM – Support Vector Machine

MAE – Mean Absolute Error

RMSE – Root Mean Square Error

MAPE – Mean Absolute Percentage Error

CSV – Comma-Separated Values

LIST OF FIGURES

CHAPTER1 : INTRODUCTION

This chapter provides a comprehensive overview of the proposed project, highlighting its importance, the motivation behind implementing it, and the main problem it seeks to address. It also discusses the project's objectives, expected outputs, and timeline, in addition to explaining the structure of the report. The purpose of this chapter is to establish a general framework that helps the reader understand the context of the project before examining the technical details presented in the subsequent chapters.

1.1 Overview

The world is witnessing increasing interest in indoor air quality, especially as individuals spend more time inside enclosed environments such as homes, offices, and classrooms. People with asthma or claustrophobia are particularly sensitive to poor indoor air quality, as they may experience breathing difficulties or anxiety attacks without having any early indicator that the environment is unsuitable.

This project focuses on developing an artificial intelligence system capable of evaluating indoor air quality using various environmental data (temperature, humidity, CO_2 level, number of occupants, and room size). The system provides an instant air quality rating, along with smart recommendations aimed at improving the indoor environment.

1.2 Project Motivation

The main motivation behind this project is the absence of a practical and fast method for evaluating indoor air quality before or during exposure, especially for individuals who are at higher health or psychological risk such as asthma patients or those with claustrophobia. Currently, people rely on personal perception or subjective experience to estimate the suitability of an indoor space, which is inaccurate and may lead to worsening symptoms.

Modern artificial intelligence technologies allow the analysis of multisource data to generate accurate assessments and predict air quality before actual symptoms appear. This enables the development of a system that supports a sensitive segment of society and enhances safety within indoor environments.

1.3 Problem Statement

People with asthma or claustrophobia struggle to determine whether the environment around them is safe or suitable before experiencing symptoms. There is no smart and quick method to identify indoor air quality, and the factors affecting air quality are multiple and constantly changing, such as humidity, CO_2 concentration, temperature, and the number of people in the room.

Therefore, the core problem can be described as follows:

There is no data-driven system capable of evaluating indoor air quality in real time and providing suitable recommendations for individuals at higher risk.

1.4 Project Aim and Objectives

Aim

To develop an artificial intelligence system capable of evaluating indoor air quality and alerting individuals who are vulnerable to asthma or claustrophobia, while providing intelligent recommendations to improve the indoor environment.

Objectives

1. Collect environmental data including temperature, humidity, CO_2 level, number of occupants, and room size.

2. Process and analyze the data to identify the key factors affecting air quality.

3. Build an AI model capable of classifying air quality (Excellent / Fair / Poor).

4. Integrate a real sensor (such as humidity or temperature) to create a practical, usable system.

5. Generate automated smart recommendations such as “Open the window” or “Reduce the number of occupants.”

6. Evaluate the model using unseen test data to verify its generalization capability.

7. Develop an initial prototype that can later be expanded into a mobile application or a physical device.

Answers to the required questions:

Q1. What is the goal the project wants to achieve?

The goal is to build an AI system capable of evaluating indoor air quality in real time and warning or guiding individuals who are at higher health or psychological risk.

Q2 . How will the project achieve this goal?

By collecting environmental data, training an artificial intelligence model to classify air quality, linking the model to real sensor readings for real-time assessment, and providing practical recommendations to improve air quality.

1.5 Project Limitations

Several limitations may affect the performance of the project, including:

1. Data limitations: Real-world indoor air quality data may be limited, so simulated data or publicly available datasets will be used for training.

2. Number of sensors used: The system currently relies on only one physical sensor (e.g., humidity or temperature), which may reduce accuracy compared to commercial multi-sensor systems.

3. Environmental variability: Air quality changes rapidly, so the system may require continuous updates or additional data to maintain accuracy.

1.6 Project Expected Output

The project is expected to produce the following outputs:

1. A trained artificial intelligence model for classifying air quality.

2. Real-time air quality assessment using the sensor's input.

3. Smart recommendations for improving indoor air quality.

4. Data analysis reports showing the most influential factors.

5. An initial prototype that can later be expanded.

6. A simple dashboard or interface displaying the results.

1.7 Project Schedule

Phase	Description	Status
Phase 1	Data Collection and Dataset Selection	Completed
Phase 2	Data Understanding and Exploration	Completed
Phase 3	Data Preprocessing and Cleaning	Completed
Phase 4	Data Integration and Transformation	Completed
Phase 5	Feature Engineering and Labeling	Completed
Phase 6	Ethical Considerations and Data Privacy Assessment	Completed
Future Phase	Model Development and Evaluation	Planned

1.8 Report Organization

Chapter 1: Introduction, motivation, problem description, objectives, limitations, and expected outputs.

Chapter 2: Literature Review

Chapter 3: Requirement Engineering and Analysis

Chapter 4: Model Development and Architectural Design

Chapter 5: Model/System Evaluation and Testing Plan

Chapter 6: Model Deployment and Integration

Chapter 7: Conclusion and future work

CHAPTER 2: LITERATURE REVIEW

We explore the existing research on indoor and outdoor air quality prediction using artificial intelligence by reviewing relevant studies. We summarize the key findings and methodologies applied in previous works, including machine learning and deep learning approaches, to provide context for our project. By examining the literature, we discuss the accuracy rates and performance metrics achieved in these studies, offering insight into the effectiveness of various models and techniques for air quality forecasting. In addition, we highlight gaps in the current knowledge and demonstrate how our research builds on and contributes to the existing work in this field, particularly in enhancing prediction reliability and supporting proactive indoor environmental management.

1.This study addresses indoor air quality (IAQ) prediction by focusing on CO2 concentrations in buildings in Shanghai from November 2016 to March 2017. Artificial intelligence models, including Machine Learning (ML) and Deep Learning (DL) techniques, were applied to sensor-collected datasets to forecast CO2 levels. Specifically, Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks were implemented to capture temporal patterns in the data. Model performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Median Absolute Error, and Coefficient of Determination (R^2). Results showed that LSTM and GRU provided the highest prediction accuracy, highlighting their suitability for time-series forecasting of IAQ. The study demonstrates the importance of selecting models tailored to dataset characteristics, enabling more reliable and efficient indoor air quality management, and supporting intelligent ventilation and occupant safety.

2.Machine Learning (ML) has been used in environmental research to predict indoor air quality, but few studies focused on PM2.5, fine particles that can reach deep into the lungs and cause respiratory and heart problems. Estimating indoor PM2.5 is important, especially for the elderly and people with health issues who spend most of their time indoors. This study used ML models including GLM with Lasso, Random Forest (RF), and XGBoost to predict six-hourly indoor PM2.5 in six non-smoking homes in Malta and Gozo, using outdoor PM and weather data. PM was measured continuously with aerosol spectrometers. Models were tested with repeated

10-fold cross-validation and hyperparameter tuning using RMSE. For homes with low indoor PM, GLM worked well but showed some correlation in results, while RF and XGBoost performed very well (IOA 0.92–0.93), with outdoor PM1 as the most important factor. When including a home with high PM from cooking, RF gave the best performance (RMSE 30.65 $\mu g/m^3$, IOA 0.66) and indoor humidity helped predict high PM. The study showed outdoor PM1 strongly affects indoor PM2.5 where indoor sources are low, while cooking can raise indoor PM2.5 up to 3.6 times above WHO guidelines, so kitchen ventilation is important to reduce exposure.

3.Indoor exposure to fine particles (PM2.5) poses health risks, prompting the use of low-cost sensors for monitoring, but their accuracy is affected by environmental conditions and instrument drift. This study introduced an automated machine learning (AutoML) calibration framework to improve the reliability of low-cost indoor PM2.5 sensors. The multi-stage framework linked low-cost field sensors with drift-correction reference sensors and a high-grade reference instrument, applying separate calibration models for low (clean air) and high (pollution events) concentrations. The system was tested in a controlled indoor chamber with two sensor models exposed to various indoor pollution sources under natural ambient conditions. AutoML calibration significantly enhanced sensor performance, achieving strong correlation with reference measurements ($R^2 > 0.90$) and substantially reducing error metrics, with RMSE and MAE roughly halved compared to uncalibrated data, while bias was minimized. This study demonstrates that AutoML calibration can convert low-cost sensors into reliable tools for indoor air quality monitoring, supporting more accurate exposure assessment and informing public health interventions.

4.Indoor exposure to bioaerosols, especially in crowded and poorly ventilated spaces, poses significant public health risks, yet real-time monitoring and near-future prediction of airborne biological particles remains challenging. This study developed artificial intelligence (AI) models using data from indoor air quality sensors and ultraviolet light-induced fluorescence observations to estimate concentrations of bioaerosols (bacteria-, fungi-, and pollen-like particles) and particulate matter (PM2.5 and PM10) in real-time and up to 60 minutes ahead. Seven AI models were trained and evaluated using data from an occupied commercial office and a shopping mall. The Long Short-Term Memory (LSTM) model achieved the highest prediction accuracy, approximately 60–80% for bioaerosols and around 90% for PM on testing and time series datasets. This study demonstrates that AI-based prediction can provide building operators with near real-time forecasts of indoor bioaerosol and PM concentrations, enabling better management of indoor environmental quality.

5.Monitoring indoor CO2 levels is essential for health, as elevated CO2 can cause headaches, fatigue, and impaired cognitive function. This study developed an adaptive IoT-based forecasting system for homes that uses low-cost wireless environmental sensors and a Smartex wearable t-shirt to monitor CO2, temperature, humidity, and occupant physical activity in real time. The model employs a Generative Adversarial Network (GAN) with LSTM networks as generator and discriminator to predict near-future CO2 values. Only five days of real data were used for training, while GAN-generated data simulated an additional five days, effectively creating a 10-day dataset. The system was tested in a residential room with one occupant performing smart work and exercise. Results showed high accuracy, with a Root Mean Square Error (RMSE) of approximately 8 ppm, and inclusion of activity data significantly improved predictions (RMSE decreased from 80.22 ppm to 7.78 ppm). This approach allows accurate, multi-step CO2 forecasting with limited data, supporting proactive ventilation control, energy efficiency, and occupant comfort in real-world indoor environments.

6.This study presents a compact smart air purifier integrating a dual-stage filtration system (Intense Field Dielectric (IFD) for PM removal and zeolite-based filter for VOC adsorption) with high-precision sensors and machine learning (ML)-based predictive control. Computational Fluid Dynamics (CFD) simulations optimized airflow and filter design. The purifier operates automatically based on real-time air quality (AQI), achieving up to 30% energy savings. Experimental results showed a 76% reduction in PM2.5 and 65% reduction in VOCs within 30 minutes. Eleven ML regression models were tested for AQI prediction, with Lasso Regression performing best ($R^2 = 0.9179$, RMSE = 5.14, MAE = 4.13). Feature analysis identified PM2.5, PM10, noise, and humidity as key predictors. The system demonstrates how combining advanced filtration with ML forecasting enables anticipatory, energy-efficient indoor air purification suitable for homes, offices, and other enclosed environments**.**

7.The study assessed air quality in Kuwait using data from 12 monitoring stations (Jan 2019–Nov 2021), including PM2.5, PM10, gases, and meteorological parameters. PM2.5 often exceeded “Good” and “Moderate” levels, especially in industrial and high-traffic areas, with “Unhealthy” peaks during summer and dust storms. Severe pollution correlated with temperatures of 15–45°C, humidity <50%, wind speeds <12 m/s, and wind directions 240–340°, showing that dry, stagnant air and winds from industrial zones worsen pollution. Seasonal peaks occurred in summer and spring. Eight ML algorithms were tested across five feature sets; Gradient Boosting and AdaBoost performed best ($R^2 \approx 0.999$) when key PM2.5/PM10 and meteorological features were included, while pruning irrelevant features improved efficiency. Tree-based ensemble methods captured complex non-linear relationships, making deep learning unnecessary. Main pollution drivers were industrial proximity, meteorology, and PM levels. Policy recommendations include reducing gas flaring, adopting renewables, promoting EVs, and importing cleaner electricity. The study demonstrates that combining environmental data, meteorological factors, and targeted ML feature selection enables accurate air quality prediction and informs effective public health and urban planning strategies.

8.Air pollution is a critical public health challenge in megacities, and traditional numerical or single-site ML models face limitations such as high computational costs and low efficiency. To address this, the FuXi-Air model was developed using multimodal data fusion, integrating meteorological forecasts, emission inventories, and pollutant monitoring data guided by air pollution mechanisms. Utilizing an autoregressive prediction framework combined with frame interpolation, FuXi-Air provides accurate 72-hour forecasts for six major pollutants at hourly resolution across multiple monitoring sites within 25–30 seconds. Ablation studies show that meteorological data contribute most to accuracy, but integrating all data sources significantly improves predictions under varying pollution mechanisms. The model demonstrates superior computational efficiency and forecasting precision compared to traditional models, offering a practical example for deploying multimodal AI-based forecasting systems and supporting smart city air pollution risk management.

9. This study addresses the critical problem of air pollution and its impact on public health by proposing an advanced machine learning approach for predicting PM2.5 concentrations. The core idea is to improve prediction accuracy by combining multiple machine learning models rather than relying on a single model. Air quality datasets collected from major cities such as Beijing and Istanbul were used, including historical PM2.5 measurements and related environmental data. Several widely used machine learning models, namely Multi-Layer Perceptron (MLP), Support Vector Regression (SVR), and Random Forest (RF), were first trained independently to learn different patterns within the data. The predictions generated by these base models were then integrated using a stacking ensemble framework, where a meta-learner combines and optimizes the outputs of all models to produce the final PM2.5 prediction. Model performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). The stacking ensemble achieved the best performance, with MAE = 6.67, RMSE = 8.80, and $R^2 = 0.91$, outperforming all individual models. These results demonstrate that the stacking ensemble approach provides higher accuracy and better generalization capability for PM2.5 prediction compared to traditional machine learning methods.

10. This study proposes an occupant-centric framework for assessing Indoor Environmental Quality (IEQ) by integrating environmental measurements with self-reported comfort and health data. Unlike traditional approaches that focus on single comfort indicators, the proposed method adopts a multimodal data fusion strategy using Transformer-based models to capture complex temporal patterns and relationships across diverse data sources. The framework was evaluated in public indoor spaces and demonstrated high classification performance, achieving 97% accuracy for comfort-based IEQ classification and 96% accuracy for health-based classification, outperforming baseline models. The results highlight the effectiveness of Transformer-based multimodal learning in providing a comprehensive and accurate assessment of indoor environmental conditions.

11. This research presents an image-based deep learning approach for predicting air quality, focusing on PM2.5 concentration. The proposed model combines a three-dimensional convolutional neural network (3D-CNN) with a gated recurrent unit (GRU) and an attention mechanism. The 3D-CNN extracts spatial and temporal features from multi-dimensional image datasets, while the GRU captures temporal dependencies efficiently. The attention mechanism adjusts the influence of features, improving prediction accuracy and reducing random fluctuations. The model was tested using images from Shanghai, pre-processed to remove noise and normalized, with data split into training and testing sets. Results show that the 3D-CNN-GRU with attention outperforms traditional CNN-LSTM and other baseline models, achieving a mean absolute percentage error (MAPE) of 15.6% and MASE of 21.57 for PM2.5. The model provides accurate multi-horizon forecasts, enabling near real-time air quality monitoring and early warning for pollution. This approach demonstrates the potential of deep learning and image-based data to improve urban air quality prediction.

12. Air pollution is a global problem that threatens environmental sustainability and public health. This study focuses on the long-term prediction of daily Air Quality Index (AQI) in eastern Türkiye from 2016 to 2024, using a dataset that includes four major air pollutants (PM_{10}, SO_2, NO_2, O_3) and five meteorological variables (temperature, precipitation, relative humidity, wind direction, wind speed). Three machine learning models—XGBoost, LightGBM, and Support Vector Machine (SVM)—were implemented to forecast AQI levels. The methodology involved preprocessing the data, splitting it into training and testing sets, and training each model to capture the relationships between environmental predictors and AQI values. Model performance was evaluated using the coefficient of determination (R^2), root mean square error (RMSE), and mean absolute error (MAE). Among the three models, XGBoost achieved the highest prediction accuracy with $R^2 = 0.999$, RMSE = 0.234, and MAE = 0.158, outperforming LightGBM and SVM. These results demonstrate that ensemble-based machine learning approaches like XGBoost can effectively model AQI fluctuations using environmental predictors, providing valuable insights for air quality forecasting systems, early warning mechanisms, public health protection, and environmental policy-making**.**

13. This study focuses on predicting the Air Quality Index (AQI) using multiple machine learning approaches, including K-Nearest Neighbors (KNN), XGBoost, and a Neural Network (NN). The dataset includes daily measurements of major air pollutants (PM2.5, PM10, NO_2, SO_2, O_3) along with meteorological variables such as temperature, humidity, wind speed, and wind direction. The methodology involved preprocessing the data, normalizing features, and splitting the dataset into training and testing sets. The KNN model was applied to forecast AQI values by identifying similar historical patterns in the data. Results showed that KNN achieved an accuracy of 85.39%, performing better than XGBoost (72.91%) but slightly lower than the Neural Network model, which achieved 98.17% accuracy. The study demonstrates that KNN is effective for capturing local patterns in environmental data and provides a reliable baseline for air quality prediction, especially in short-term forecasting scenarios**.**

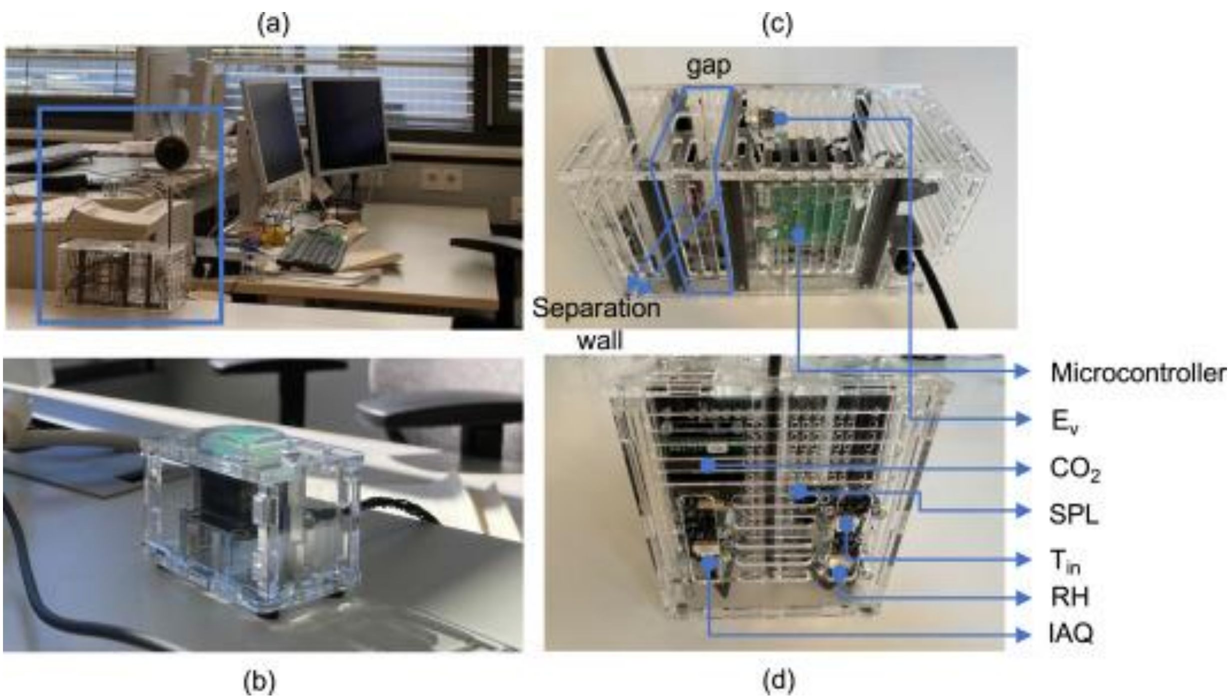
Data Collection Methods and Sources

The data used in this project was obtained from a publicly available dataset titled *"Non-intrusive indoor environmental data in two double-occupied offices"*. The measurements were collected as part of a longitudinal study conducted in two naturally ventilated offices at the Munich University of Applied Sciences in Germany. Each office is equipped with manually operable windows, exterior shading, and a heating system.

An IoT sensory device was installed at desk level near the occupants to continuously monitor the indoor environment. The sensors recorded several physical variables, including **air temperature, CO_2 concentration, indoor air quality (IAQ) index, illuminance, relative humidity, sound pressure level, and air speed, alongside the window status**. Occupancy ground truth was recorded as a **binary label** (occupied vs. unoccupied), while the window state (open/closed) was transmitted via an EnOcean-based contact sensor.

The IAQ index in this dataset is defined on a scale from 0 to 500, where smaller values indicate better indoor air quality and larger values represent increasingly degraded conditions. Internally, the index is derived from the concentration of multiple volatile organic compounds, providing a compact indicator of overall indoor air pollution.

Figure1



The raw data is distributed as two CSV files, **office_A.csv** and **office_B.csv**, each corresponding to one of the monitored offices. Both files were downloaded from the Mendeley Data repository and form the starting point for the preprocessing and analysis steps described in the following subsections.

Why We Merged the Two Datasets (office_A and office_B)

Both ***office_A*** and ***office_B*** contain the same structure, sensor measurements, and data collection methodology. Therefore, merging the two files into a single unified dataset was an essential step for several reasons:

First, merging increases the overall dataset size, which enhances the model's ability to generalize and improves prediction accuracy.
Second, combining data from two different office environments introduces more variability in environmental conditions and occupant behavior, making the dataset more representative.
Third, working with one consolidated dataset simplifies preprocessing, analysis, and model development.

Figure 2

```
Same columns (same order)? True
Columns only in A: []
Columns only in B: []
Same column names (ignoring order)? True
```
:

Verifying Column Consistency Before Merging

Before merging the two datasets (*office_A* and *office_B*), it was essential to verify that they shared the same column names and structure. Ensuring column consistency prevents alignment issues, incorrect data mapping, and potential loss of information during the merge operation. Therefore, we checked whether both files contained identical column names, in the same order and with no missing or additional fields. The verification results confirmed that the datasets matched perfectly, allowing us to safely proceed with merging them into a unified dataset.

Figure 3

```
All common columns have the same dtypes in A and B ✅
```

Before merging the two datasets (**office_A** and **office_B**), it was crucial to ensure that both files used the same data types for all shared columns. Differences in data types—such as one file storing a column as *float* while the other stores it as *int*—can lead to merge errors, incorrect calculations, or unexpected model behavior later in the workflow.

To avoid these issues, we performed a **data type alignment check** on all common columns across the two datasets. The results showed that **all shared columns have identical data types in both files**, confirming that the datasets are structurally compatible and safe to merge into a unified dataset.

This verification step ensures clean integration and prevents downstream inconsistencies during preprocessing and model development.

Figure 4

```
Saved: office_all.csv
Shape: (144000, 12)
```

After verifying the column consistency and data types, the two raw files (office_A.csv and office_B.csv) were concatenated into a single integrated dataset named office_all.csv. The resulting dataset contains **144,000 rows and 12 columns**, providing a large number of indoor environmental measurements for subsequent analysis and modelling.

Figure5

	Unnamed: 0	DAY	TIME	MONTH	OCCUPANCY	WINDOW	Ev	IAQ	CO2	SPL	RH	Tin
0	0	1	0:00:00	September	0	0	0.01	72.40	404.25	27.86	53.45	26.18
1	1	1	0:01:00	September	0	0	0.01	73.05	404.07	27.88	53.45	26.18
2	2	1	0:02:00	September	0	0	0.01	74.08	404.07	27.79	53.45	26.18
3	3	1	0:03:00	September	0	0	0.01	72.27	406.42	27.77	53.42	26.18
4	4	1	0:04:00	September	0	0	0.01	73.98	407.35	27.79	53.45	26.17

During preprocessing, several columns were removed because they did not contribute meaningful information or were redundant for the machine learning workflow. These include:

---

1. Unnamed: 0

An automatically generated index column that does not contain any real sensor information.
It was removed as it has no analytical value.

---

2. DAY and TIME

These timestamp components do not provide useful predictive information for the indoor air quality task.
Since the project does not involve time-series modeling, keeping them would only introduce unnecessary noise.

---

3. Ev

This column had a constant value (~0.01) across the entire dataset.
Constant features do not contribute to model learning and are therefore removed.

Figure6

	MONTH	OCCUPANCY	WINDOW	IAQ	CO2	RH	Tin
0	September	0	0	72.40	404.25	53.45	26.18
1	September	0	0	73.05	404.07	53.45	26.18
2	September	0	0	74.08	404.07	53.45	26.18
3	September	0	0	72.27	406.42	53.42	26.18
4	September	0	0	73.98	407.35	53.45	26.17

Data Type Standardization

As part of the preprocessing phase, we examined the data types of all columns to ensure consistency and compatibility with the subsequent analysis steps. During this inspection, we found that all variables were already represented in numeric format except for the MONTH column, which was stored as a text value (e.g., "September"). Since non-numeric values can cause issues during analysis and cannot be directly used in machine learning workflows, the column was transformed into a numerical representation called **MONTH_NUM** reflecting the actual month number.

It is important to note that this transformation **does not constitute an encoding technique** such as label encoding or one-hot encoding. The month variable was not treated as a categorical label; instead, it was converted into a natural numeric form because months inherently follow an ordered temporal structure. Therefore, this step falls under **data standardization**, not encoding.

After this conversion, all columns in the dataset were fully numeric, ensuring that the data is clean, consistent, and ready for the next stages of preprocessing and modeling.

Figure7

	OCCUPANCY	WINDOW	IAQ	CO2	RH	Tin	MONTH_NUM
0	0	0	72.40	404.25	53.45	26.18	9
1	0	0	73.05	404.07	53.45	26.18	9
2	0	0	74.08	404.07	53.45	26.18	9
3	0	0	72.27	406.42	53.42	26.18	9
4	0	0	73.98	407.35	53.45	26.17	9

Figure8

	0
OCCUPANCY	0
WINDOW	0
IAQ	0
CO2	0
RH	0
Tin	0
MONTH_NUM	0

dtype: int64

Missing Values Analysis

we examined the dataset for missing values. For this purpose, we computed the number of null entries in each column using the **isna().sum()** function. The results showed that all features (OCCUPANCY, WINDOW, IAQ, CO2, RH, Tin, and MONTH_NUM) contained **zero missing values.**

Since no null or undefined entries were present in the integrated dataset, no imputation or row removal was required at this stage. This confirms that the data is complete and ready for subsequent preprocessing and analysis steps.

Outlier Detection Using the IQR Method

To ensure the reliability of the minute-level sensor data, we performed outlier detection using the Interquartile Range (IQR) method. This technique was selected because environmental sensor readings (such as CO_2, humidity, temperature, and IAQ) naturally fluctuate throughout the day, and not every extreme value should be treated as an error.

The IQR method is particularly suitable for this type of data for several reasons:

1. **It is robust to extreme values** and does not rely on the mean, making it ideal for skewed or non-normally distributed sensor data.

2. **It avoids misclassifying natural fluctuations as anomalies**, which is essential when working with high-resolution indoor environmental measurements.

3. **It is widely used in scientific literature** for analyzing environmental and IoT sensor datasets due to its simplicity and stability.

4. Importantly, in our workflow, the IQR method was applied **only to identify outliers**, not to remove or modify them, ensuring that the raw environmental patterns remain intact for accurate modeling.

The analysis revealed outliers in a few numerical features (e.g., CO_2, RH, Tin, IAQ), which helped us understand the variability and noise level in the dataset without compromising its integrity.

Figure9

```
Outliers per column (IQR):
{'CO2': 2089, 'RH': 40, 'Tin': 5873, 'IAQ': 17}
```

IMPORTENT

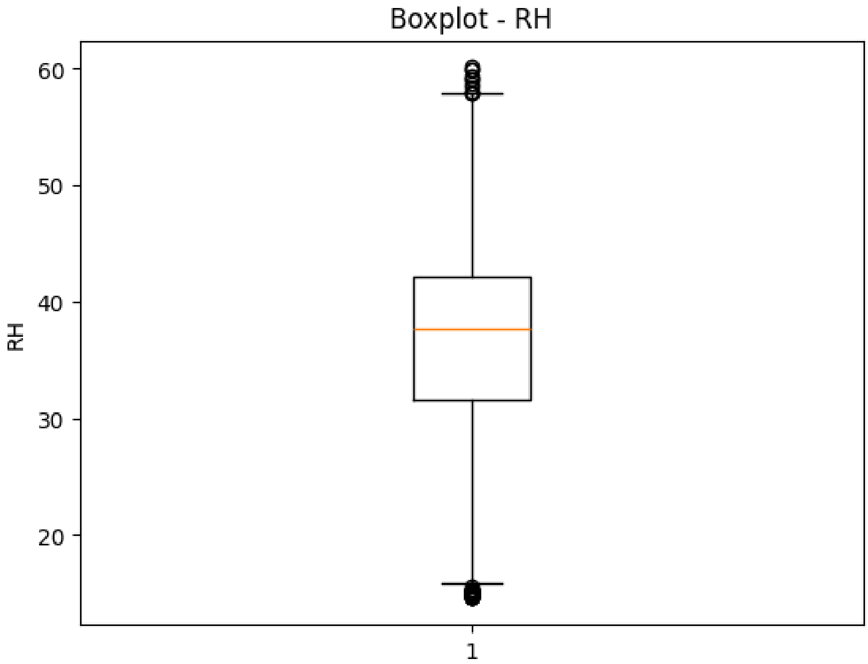
Why Outliers Were Detected but Not Removed

During the preprocessing stage, we applied the Interquartile Range (IQR) method to identify potential outliers in key environmental variables such as CO_2, IAQ, temperature, and relative humidity. However, although outliers were detected, **we intentionally chose not to remove them**. This decision is critical for maintaining the integrity and realism of the dataset.

In indoor environmental monitoring, extreme or unusual readings are not necessarily sensor errors—they often represent **real and meaningful environmental events**, such as sudden increases in occupancy, poor ventilation, window closure, or conditions that may trigger discomfort or health risks for sensitive individuals (e.g., asthma patients or people with claustrophobia).
Such rare but important events reflect high-risk scenarios that the predictive model must learn to recognize. Removing them would eliminate essential patterns related to unsafe or abnormal environmental conditions and would lead to a model that performs well only under normal circumstances but fails when detecting critical deviations.

For these reasons, IQR-based outlier detection was used **solely for analysis and understanding the behavior of the data**, not for filtering or modifying it. All values were retained to ensure that the dataset remains realistic, representative, and capable of supporting a robust machine learning model that can generalize to both typical and extreme environmental situations.

Figure10



Boxplot Visualization for Outlier Identification

To visually inspect the presence of outliers in the environmental features, we used boxplots. The figure above shows an example boxplot for the **relative humidity (RH)** feature. In a boxplot, the central box represents the interquartile range (IQR), which spans from the 25th percentile (Q1) to the 75th percentile (Q3). The horizontal line inside the box indicates the median value of the distribution.

The “whiskers” extend from the box to the lowest and highest values that fall within an acceptable range, typically defined as **1.5 × IQR** below Q1 and above Q3. Any data points plotted as isolated markers beyond the whiskers are considered **potential outliers**.

In the RH boxplot, the scattered points above the upper whisker and below the lower whisker represent readings that lie outside the normal humidity range observed in the office environment. This graphical technique complements the IQR-based numerical analysis by providing an intuitive view of how extreme values are distributed relative to the bulk of the data.

```
np.int64(6)
```

When checking for duplicate rows using **df.duplicated().sum(),** a small number of records appeared as duplicates based on the sensor values only (e.g., CO_2, RH, Tin, IAQ). However, closer inspection showed that these rows did **not** share the same timestamp. In other words, the environmental measurements were identical or very similar, but they were recorded at **different time points**.

Such repetitions are expected in indoor environmental data, where conditions can remain stable over several minutes. Therefore, these rows do not represent true duplicates of the same event, but rather valid observations of similar conditions occurring at different times. For this reason, we chose **not to remove** them, and all rows were retained to preserve the full temporal structure of the dataset.

Noise / Validity Checks for Sensor Data

To ensure that the dataset does not contain unrealistic or noisy measurements, we applied a series of rule-based validity checks for each feature, based on the sensor descriptions in the paper:

- **OCCUPANCY (0 or 1)**
  The OCCUPANCY column represents whether the office is occupied or not. We verified that all values are either 0 (no occupants) or 1 (at least one occupant). Any value not in {0, 1} would be flagged as invalid. The check returned 0 invalid values, confirming consistent binary encoding.
- **WINDOW (0 or 1)**
  The WINDOW feature encodes the window state. Similar to occupancy, we enforced a binary domain with 0 = closed and 1 = open. Values outside this set are considered noisy or mis-recorded. No violations were detected for this column.

- **Relative Humidity RH ([0, 100])**

  For RH, which represents relative humidity in percent, we constrained the readings to the physically meaningful range [0, 100]. Any value below 0% or above 100% was treated as implausible and thus noise. The validation showed that all humidity values lie within this range (0 invalid readings).

- **CO_2 concentration (> 0)**

  The CO2 column stores the indoor carbon dioxide concentration in ppm. Since CO_2 cannot be zero or negative in a real environment, we required all values to be strictly greater than 0. The check confirmed that there are no non-positive CO_2 values in the dataset.

- **Indoor Air Quality Index IAQ ([0, 500])**

  The IAQ feature is an indoor air quality index defined in the original study, with a theoretical range from 0 (good air quality) to 500 (very poor air quality). We therefore verified that all IAQ values fall within the interval [0, 500]. Values outside this range would indicate sensor or processing errors. The noise check returned 0 out-of-range IAQ values, suggesting that the recorded index values are consistent with the expected specification.

Overall, all these checks reported zero invalid entries, which indicates that the dataset is free from obvious noise or corrupted measurements according to our domain-based validation rules.

Feature Engineering and Selection

CO2–Occupancy Interaction Feature (CO2_x_OCCUPANCY)

Figure11

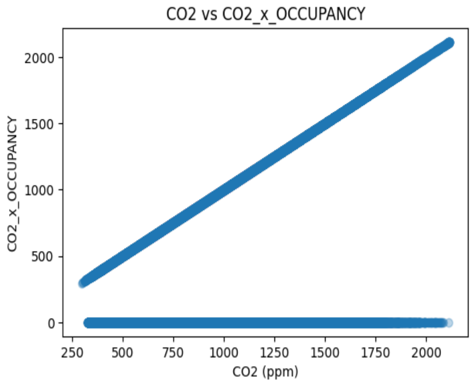
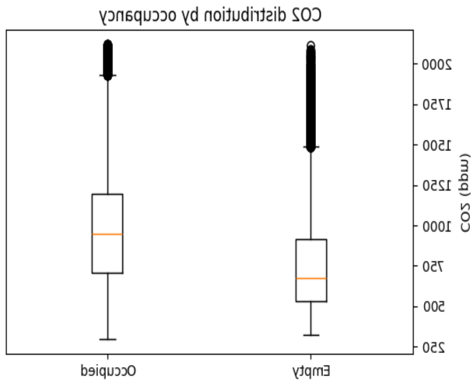


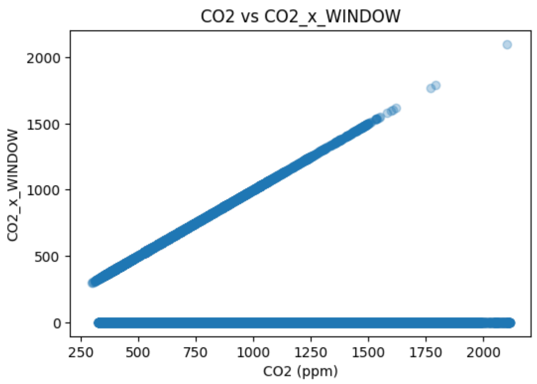
Figure12



To capture how indoor CO_2 levels relate specifically to human presence, we created an interaction feature by multiplying the CO_2 concentration with the binary occupancy indicator (0/1). When the office is empty, this feature is zero, but when occupants are present it equals the measured CO_2 value. This transformation helps the model distinguish between “background” CO_2 fluctuations and CO_2 that is directly associated with human activity, which is particularly relevant for people with asthma or sensitivity to poorly ventilated spaces. By explicitly encoding this interaction, the model can better learn patterns where high CO_2 levels *while occupied* indicate degraded air quality and potential discomfort, rather than treating all CO_2 readings as equally important.

CO2–Window Interaction Feature (CO2_x_WINDOW)

Figure13



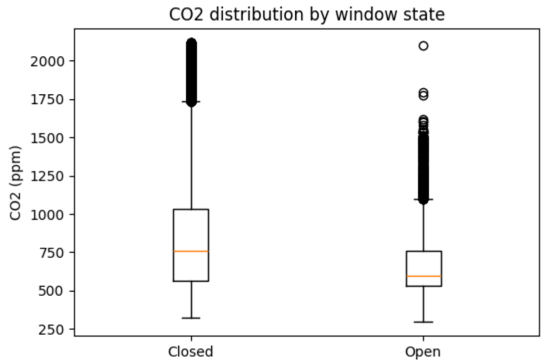


Figure14

We also engineered an interaction feature between CO_2 concentration and the binary window state. The **CO2_x_WINDOW** feature is equal to the measured CO_2 value when the window is open and zero otherwise. This feature allows the model to explicitly capture how effective natural ventilation is in reducing CO_2 levels. For example, high CO_2 values when the window is open may indicate insufficient airflow or overcrowding, while high CO_2 with the window closed is expected but still relevant for risk assessment. Encoding this interaction helps the model distinguish between "ventilated" and "non-ventilated" high-CO_2 situations, which is important for generating meaningful recommendations such as opening a window or reducing occupancy.

Combined Window–Occupancy Indicator (WINDOW_x_OCCUPANCY)

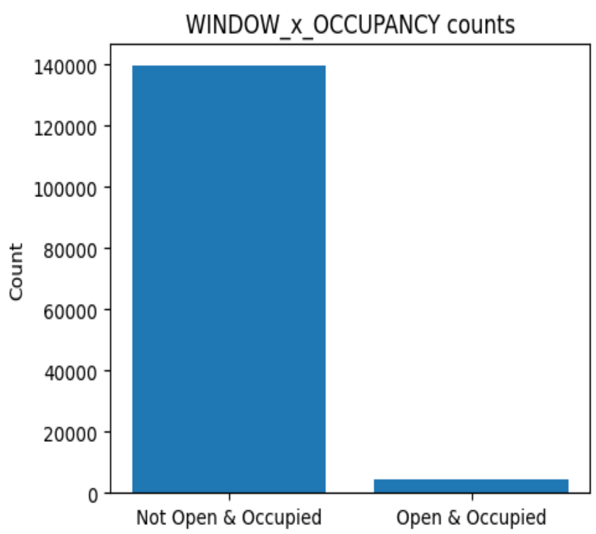
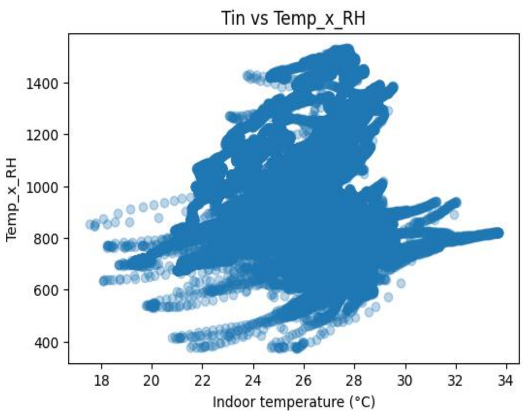
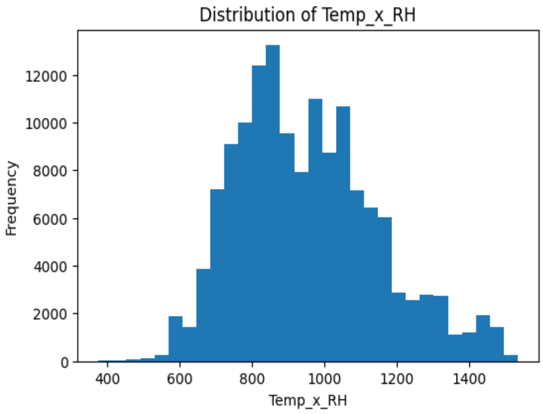


Figure15

To explicitly represent situations where occupants are present *and* the window is open, we created a binary interaction feature by multiplying the window state (0/1) with the occupancy indicator (0/1). The resulting **WINDOW_x_OCCUPANCY** feature equals 1 only when people are in the office and the window is open; in all other cases it is 0. This feature is useful because it directly captures whether ventilation is being used while the room is occupied, which is critical for maintaining healthy air quality. It also simplifies the model's reasoning: instead of learning complex patterns over two separate binary variables, the model can rely on a single, semantically meaningful indicator of "ventilated-and-occupied" conditions.

Figure16 **Figure17**





Human comfort in indoor environments is influenced not only by temperature or humidity alone, but by their combination. To capture this relationship, we engineered a composite feature Temp_x_RH by multiplying indoor temperature (Tin) with relative humidity (RH). This simple proxy provides the model with a continuous measure that increases when both temperature and humidity are high—conditions that are often perceived as stuffy or uncomfortable and may trigger symptoms for individuals with asthma or claustrophobia. While it is not a full psychrometric comfort index, this engineered feature helps the model better detect "uncomfortable air" scenarios compared to using temperature and humidity as separate, independent inputs.

Figure18

	OCCUPANCY	WINDOW	IAQ	CO2	RH	Tin	MONTH_NUM	CO2_x_OCCUPANCY	CO2_x_WINDOW	WINDOW_x_OCCUPAN	WINDOW_x_OCCUPANCY	Temp_x_RH
0	0	0	72.40	404.25	53.45	26.18	9	0.0	0.0	0	0	1399.3210
1	0	0	73.05	404.07	53.45	26.18	9	0.0	0.0	0	0	1399.3210
2	0	0	74.08	404.07	53.45	26.18	9	0.0	0.0	0	0	1399.3210
3	0	0	72.27	406.42	53.42	26.18	9	0.0	0.0	0	0	1398.5356
4	0	0	73.98	407.35	53.45	26.17	9	0.0	0.0	0	0	1398.7865

Target Variable Definition (Labeling) Using the Indoor Air Quality (IAQ) Index

Importent

The **Indoor Air Quality (IAQ)** index was used in this project as the reference indicator (ground truth) for defining air quality categories. IAQ is not an arbitrary or subjective metric; rather, it is a **standardized composite index** computed using the **Bosch BSEC (Bosch Sensortec Environmental Cluster) algorithm**, which is widely applied with Bosch gas sensors such as the **BME680/BME688**. This algorithm converts raw gas sensor readings related to volatile organic compounds (VOCs), together with environmental compensation factors, into a normalized IAQ score ranging from **0 to 500**, where lower values indicate better air quality.

According to the official Bosch BSEC documentation, the IAQ scale is associated with qualitative air quality descriptions. Typical classifications define ranges such as *good*, *average*, *poor*, and *very poor* air quality based on specific IAQ intervals. These categories are commonly used in industrial applications, smart building systems, and indoor environmental monitoring studies.

Based on this **established and documented classification**, the IAQ values in this project were mapped into three consolidated categories to simplify interpretation while preserving their scientific meaning. Specifically, IAQ values from **0 to 100** were labeled as **Good**, values from **101 to 200** as **Moderate**, and values greater than **200** as **Poor**. This consolidation merges closely related subcategories from the original scale into broader classes that are easier to interpret and communicate, without deviating from the underlying standard.

Importantly, using IAQ as the sole basis for labeling helps avoid **data leakage**, which may occur if individual environmental variables (such as CO_2 concentration, humidity, or temperature) are directly or indirectly used to define the target classes. By relying on IAQ as an independent reference index and keeping other environmental measurements separate from the labeling process, the categorization remains methodologically sound and unbiased.

This labeling strategy ensures that the defined air quality classes are **standard-compliant, scientifically justified, and interpretable**, providing a reliable foundation for subsequent analysis and evaluation of indoor air conditions.

Figure19

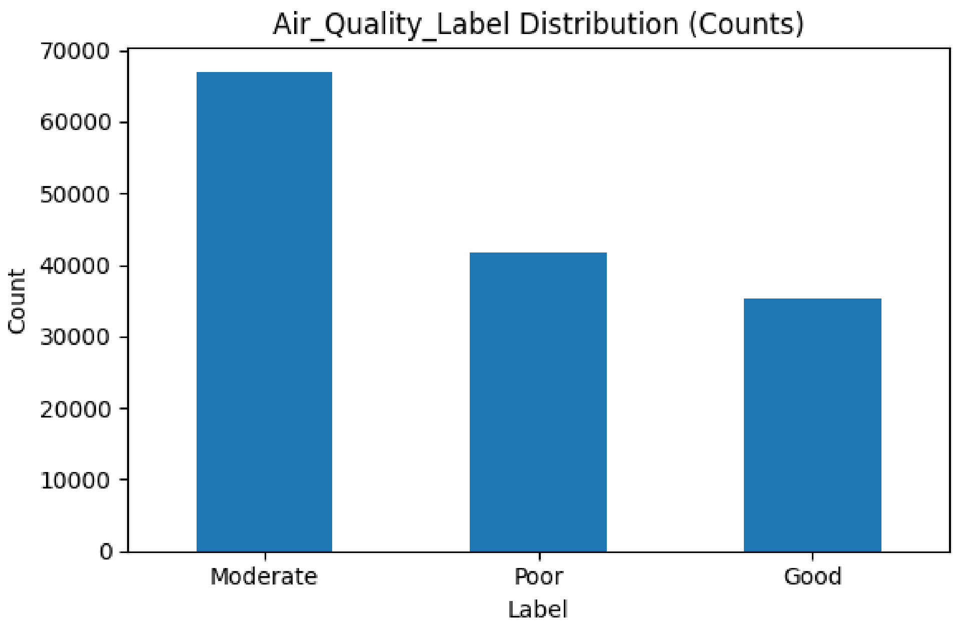
	count
Air_Quality_Label	
Moderate	66942
Poor	41740
Good	35318

dtype: int64

Figure20

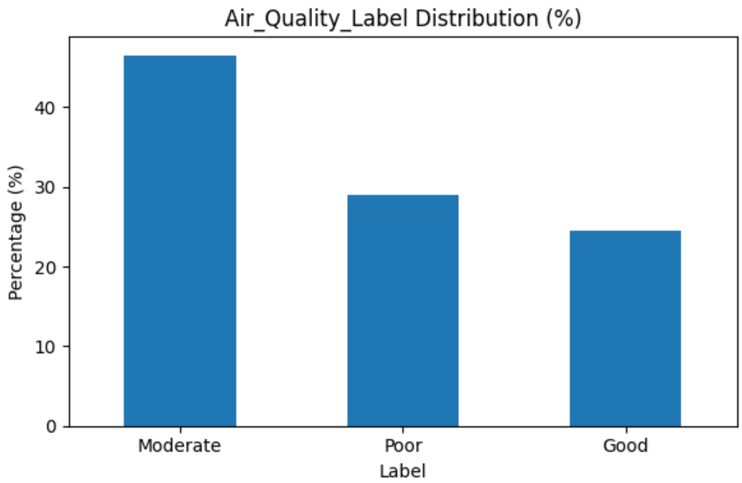
```
np.int64(0)
```

Figure21



This figure illustrates the absolute number of samples in each air quality class (Good, Moderate, Poor) after labeling based on the IAQ index. It provides an overview of the class distribution in the dataset and highlights the dominance of the Moderate air quality category.

Figure22



This bar chart presents the percentage distribution of air quality labels. Compared to the count-based representation, this figure emphasizes the relative weight of each class and supports decisions related to model evaluation and potential class imbalance handling.

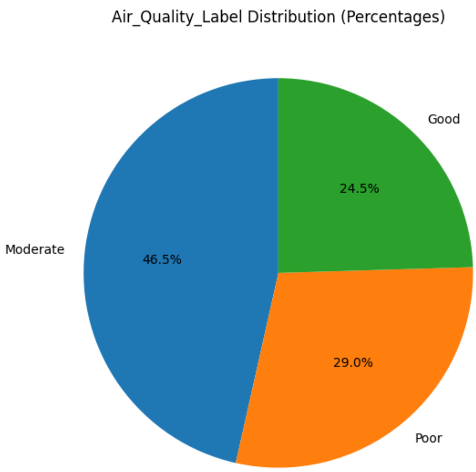
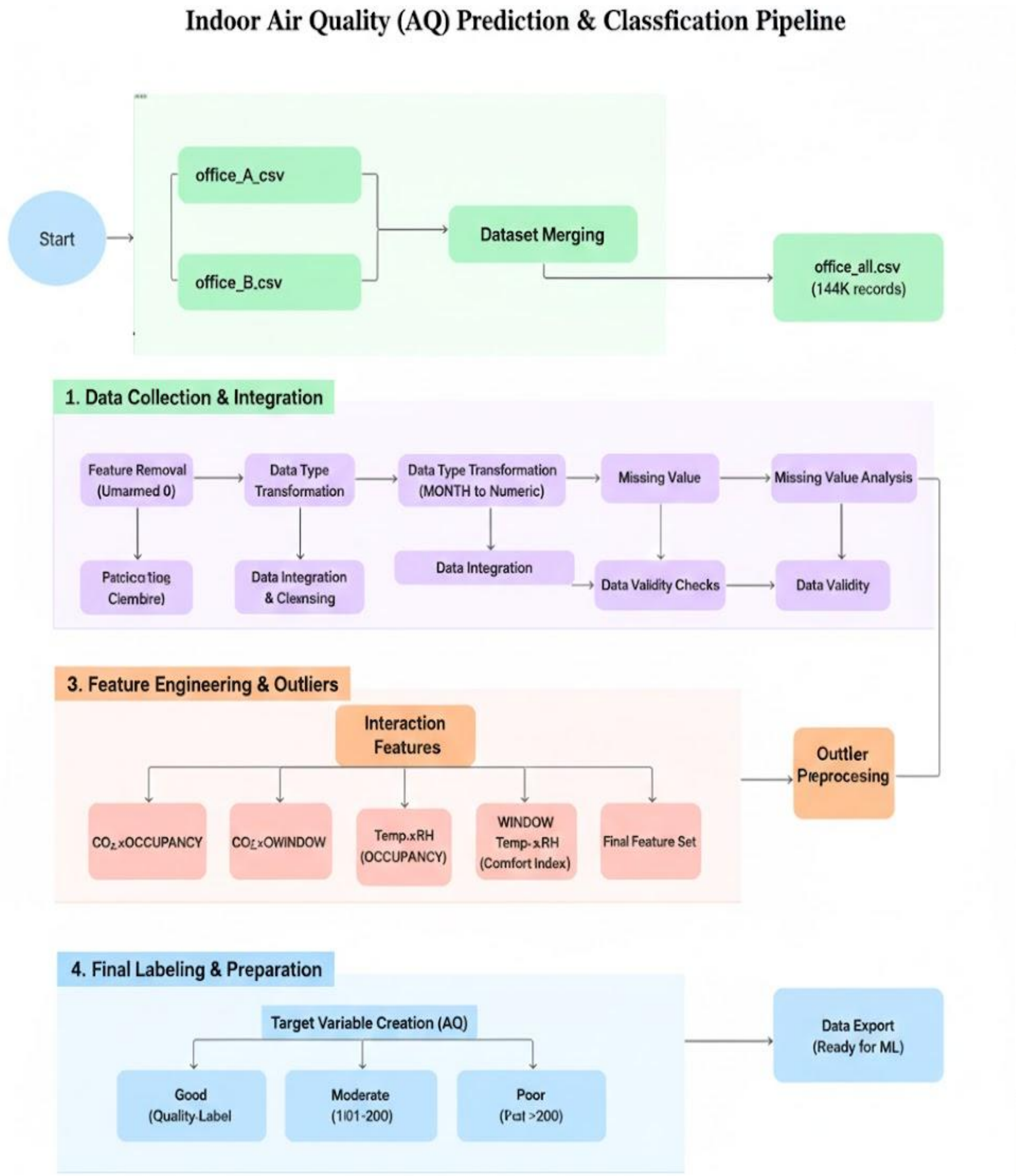


Figure23

This pie chart shows the relative percentage of each air quality label in the dataset. The visualization helps in understanding the proportional representation of Good, Moderate, and Poor air quality conditions, which is important for assessing class balance

Figure24


Indoor Air Quality (AQ) Prediction & Classfication Pipeline
Start
office_A_csv
office_B.csv
Dataset Merging
office_all.csv
(144K records)
1. Data Collection & Integration
Feature Removal
(Umarmed 0)
Data Type
Transformation
Data Type Transformation
(MONTH to Numeric)
Missing Value
Missing Value Analysis
Patcicatiog
Ciembire)
Data Integration
& Clewnsing
Data Integration
Data Validity Checks
Data Validity
3. Feature Engineering & Outliers
Interaction
Features
CO_2 xOCCUPANCY
CO xOWINDOW
Temp.xRH
(OCCUPANCY)
WINDOW
Temp-xRH
(Comfort Index)
Final Feature Set
Outller
Pieprocesing
4. Final Labeling & Preparation
Target Variable Creation (AQ)
Good
(Quality-Label
Moderate
(1101-200)
Poor
(Pat >200)
Data Export
(Ready for ML)

3.3 Ethical Considerations and Data Privacy

At this stage, the project focuses exclusively on data understanding and preprocessing, without progressing to final prediction, deployment, or real-world application. Throughout this phase, ethical considerations and data privacy principles were carefully addressed to ensure responsible and transparent handling of the data.

The dataset used in this project consists solely of non-intrusive environmental measurements, including indoor temperature, relative humidity, carbon dioxide concentration, window status, and occupancy indicators. No personally identifiable information (PII) or sensitive personal or health-related data were collected, stored, or processed at any point during this work.

The data were obtained from a publicly available research dataset intended for academic use. Issues related to informed consent were addressed at the data collection stage by the original data providers, while the current project is limited to secondary data analysis. All data were analyzed in an anonymized and aggregated form, ensuring that individual occupants cannot be identified directly or indirectly.

It is important to emphasize that this project does not aim to provide medical diagnosis or clinical recommendations. The analysis is strictly limited to assessing general indoor air quality conditions based on environmental indicators. Any potential future use of the system would be advisory and informational in nature and should not be considered a substitute for professional medical judgment or healthcare services.

Finally, the scope of this project aligns with general data protection principles and regulations such as the General Data Protection Regulation (GDPR) and the Health Insurance Portability and Accountability Act (HIPAA). As no personal or health-related data are involved, the project maintains ethical compliance, data integrity, and privacy protection throughout the preprocessing stage.

Some references are books or official standards that are not freely available as PDFs, so I provided official publisher pages or DOI links to ensure verifiability.

References

1. **Banihashemi, F., & Weber, M. (2024). *Non-intrusive indoor environmental data in two double-occupied offices* (Version 2). Mendeley Data. https://doi.org/10.17632/v7kw9ycccp.2**
2. **World Health Organization. (2010). *WHO guidelines for indoor air quality: Selected pollutants*. World Health Organization. https://www.who.int/publications/i/item/WHO-HEP-ECH-EHD-11.03**
3. **United States Environmental Protection Agency. (2023). *Introduction to indoor air quality*. https://www.epa.gov/indoor-air-quality-iaq**
4. **United States Environmental Protection Agency. (2022). *Technical overview of volatile organic compounds (VOCs)*. https://www.epa.gov/indoor-air-quality-iaq/technical-overview-volatile-organic-compounds**
5. **ASHRAE. (2019). *ASHRAE Standard 62.1: Ventilation for acceptable indoor air quality*. American Society of Heating, Refrigerating and Air-Conditioning Engineers. https://www.ashrae.org/technical-resources/standards-and-guidelines**
6. **Bosch Sensortec. (2022). *BME680 gas sensor: Measuring indoor air quality*. https://www.bosch-sensortec.com/products/environmental-sensors/gas-sensors/bme680/**
7. **Bosch Sensortec. (2022). *BSEC software: Indoor air quality index*. https://www.bosch-sensortec.com/software-tools/software/bsec/**
8. **Tukey, J. W. (1977). *Exploratory data analysis*. Addison-Wesley. https://www.jstor.org/stable/2237848**
9. **Aggarwal, C. C. (2017). *Outlier analysis* (2nd ed.). Springer. https://link.springer.com/book/10.1007/978-3-319-47578-3**

10. Chandola, V., Banerjee, A., & Ku9mar, V. (2009). Anomaly detection: A survey. *ACM Computing Surveys, 41*(3), 1–58. https://dl.acm.org/doi/10.1145/1541880.1541882
11. Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The elements of statistical learning: Data mining, inference, and prediction* (2nd ed.). Springer. https://link.springer.com/book/10.1007/978-0-387-84858-7
12. Kuhn, M., & Johnson, K. (2013). *Applied predictive modeling*. Springer. https://link.springer.com/book/10.1007/978-1-4614-6849-3
13. Kaufman, S., Rosset, S., & Perlich, C. (2012). Leakage in data mining. *ACM SIGKDD Explorations, 13*(1), 28–34. https://dl.acm.org/doi/10.1145/2382577.2382580
14. Indoor Air Quality Prediction in Sick Building Using Machine and Deep Learning: Comparative Analysis. (2025).
15. Camilleri, R., Harrison, R. M., & Aquilina, N. J. (2025). Application of Machine Learning Algorithms In Predicting Indoor Residential PM2. 5 Concentrations. Atmospheric Pollution Research,(102609).
16. Qian, J., Wynn, T., Liu, B., Shan, Y., Bartington, S. E., Pope, F. D., Dai, Y., and Shi, Z.: Enhancing Accuracy of Indoor Air Quality Sensors via Automated Machine Learning Calibration, EGUsphere [preprint], https://doi.org/10.5194/egusphere-(2025-3839, 2025).
17. (Hash MG, Forsyth A, Coleman BA, Li V, Vinagolu-Baur J, Frasier KM.Cureus (2025 Apr).
18. Leone, A., Manni, A., Caroppo, A., & Rescio, G. (2025). Adaptive IoT-Based Platform for CO_2 Forecasting Using Generative Adversarial Networks: Enhancing Indoor Air Quality Management with Minimal Data. *Engineering Proceedings*,.
19. Beldar, P., Kushare, P., Patil, A. *et al.* AI-powered AQI forecasting in an intelligent air purifier featuring intense field dielectric and Zeolite filters. *J. Eng. Appl. Sci.* 72, 150 (2025).
20. Alrashidi, H., Sibai, F. N., Abonamah, A., Alrashidi, M., & Alsaber, A. (2025). PM2.5: Air Quality Index Prediction Using Machine Learning: Evidence from Kuwait's Air Quality Monitoring Stations.
21. . (FuXi-Air: Multimodal data fusion for high-precision air quality forecasting in megacities. (2025). *arXiv preprint* arXiv:2506.07616 [cs.LG].

22. **Emeç, M., Yurtsever, M. A novel ensemble machine learning method for accurate air quality prediction. *Int. J. Environ. Sci. Technol.* 22, 459–476 (2025).**

23. **Ali, H. H., M. Abdullah, and M. Wedyan. 2022. "Application of machine learning techniques to predict patient's satisfaction of indoor environmental quality in Jordanian hospitals.**

24. **rium. (Year). *An image-based deep learning approach for multi-horizon PM2.5 forecasting using 3D-CNN-GRU with attention mechanism***

25. : Tırınk, S. (2025). Machine learning-based forecasting of air quality index under long-term environmental patterns: A comparative approach with XGBoost, LightGBM, and SVM.

26. Air Quality Index (AQI) Prediction Using Machine Learning and Deep Learning Approaches. (2025). International Journal for Research in Applied Science & Engineering Technology (IJRASET).